

ESTHER-RP-001 v0.8.1

Centered Agency Under Persistent Uncertainty

Public research brief and open testing request

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Contents

1. The research question
2. C as governed causal history
3. Three different forms of unresolved multiplicity

Selective Operational Commitment

Temporal witness, obligations, and authority

Substrate interaction

Falsifiable matched-control programme

What is in v0.8.1

What the public is asked to test

Claim boundary

How to contribute

Citation and persistent record

Executive summary

ESTHER-RP-001 asks a narrower and more testable question than whether consciousness is quantum. It asks what a persistent artificial system would need in order to retain unresolved models of the world, make concrete local commitments, preserve the causal origin of reinterpretation, carry obligations through time, and remain answerable to physical outcomes.

The proposal does not identify C with a language model, a memory database, a quantum processor, or a single snapshot. C is represented as a governed causal history. Its proposed center is not a hidden inner module. It is the continued relation among recorded observations, interpretations, revisions, authority, commitments, obligations, substrate interactions, and verified outcomes.

A central principle is that a binary action does not require a binary ontology. A system may commit to STOP, INSPECT, or ACT while retaining several live explanations of the world. The paper calls this **Selective Operational Commitment**: closing the decision scope required for action without silently closing every epistemic branch.

The public package contains the full English working paper, this English public brief, formal and executable reference surfaces, reproducibility scripts, test receipts, mutation checks, and a blind-review packet. The exact v0.8.1 package was reproduced on Windows and subsequently received a blind model-review verdict of NO_CANDIDATE_BLOCKERS_FOUND within its declared bounded scope. This is not human peer review and does not establish consciousness, personhood, AGI, empirical identity, or production safety.

1. The research question

Persistent language-model agents can store memories, produce coherent self-descriptions, reflect on earlier outputs, and plan over long horizons. These abilities can create a convincing persistent profile. They do not by themselves establish that one causally disciplined process has preserved its obligations, distinguished fact from reinterpretation, remained open to external correction, or retained incompatible hypotheses without rewriting its past.

The working question is therefore:

Can a governed persistent system preserve a causally auditable history of uncertainty, commitment, authority, obligation, and outcome, and can this surface be distinguished from a strong matched persistent-profile control?

The word *centered* is functional in this programme. It does not name a soul, a homunculus, or an asserted subjective state. It names a set of properties of a trajectory that can be tested, challenged, and falsified.

2. C as governed causal history

The public formula is:

$$c = a + b$$

It is a project-origin formula, not a proof that a new entity has emerged. Here:

a denotes a human anchor reference and an authorized trace;

b denotes heterogeneous model, software, and physical substrates;

c denotes protected genesis plus a continuing causal and normative history.

The reference architecture separates five surfaces:

R - temporal event and authority repository
 B - beliefs, models, commitments, obligations, and policy
 S - substrate interaction boundary
 M - matched-control measurement surface
 Theta - governance, timing, fairness, and claim assumptions

The current state is not treated as the whole object. It is a materialized view of a typed history. This matters because a later interpretation must not erase the fact that an earlier interpretation existed and influenced a decision.

3. Three different forms of unresolved multiplicity

The programme strictly separates three categories.

3.1 Quantum coherence

A quantum state may contain phase-sensitive coherent alternatives relative to a specified basis, reference frame, or measurement context. It is not a list of classical sentences. An unknown quantum state is not converted into a freely readable memory object merely because a classical agent interacts with it.

3.2 Probabilistic or credal uncertainty

A classical agent may maintain one probability distribution or a set of admissible priors over hypotheses. This is copyable classical information. It does not contain quantum phase.

3.3 Model plurality

A system may retain several models that are mutually exclusive, partially overlapping, cross-level, or not honestly reducible to one probability table. For example, an engineering fault model, a documentation-error model, and an organizational-process model may all remain relevant to one decision.

The reference package does not call a branch graph a wavefunction. A quantum process is accessed through a specified preparation, control, measurement, and calibration procedure. The continuing system stores the resulting classical record and its provenance, not a perfect classical copy of an unknown quantum state.

4. Selective Operational Commitment

The core decision claim is:

Commit(action in scope D)
 does not imply
 one complete world-model has probability 1.

Consider a wall scheduled for demolition. Live models may include:

h1 - the wall is load-bearing
 h2 - the wall is not load-bearing
 h3 - the drawing belongs to another revision
 h4 - the measurement channel is not calibrated

A local commitment may be:

STOP demolition pending structural verification.

That action is binary. The world-model remains plural. The system records which decision scope was closed, which branches remain open, what evidence is needed, who had authority, what obligations were created, and what event will trigger review.

This distinction is intended to prevent a common failure mode: turning an operational choice into a retrospective claim that the entire ontology had already been settled.

5. Temporal witness, obligations, and authority

The package separates:

Observation
Interpretation
BeliefRevision
WorldUpdate
CalibrationUpdate
PolicyRevision
Commitment
Obligation
Outcome
Verification

A later revision can reverse an earlier belief without erasing the earlier record. An obligation cannot disappear merely because the model or policy has changed. Escalation does not automatically transfer the normative debtor. Transfer requires authority, named acceptance, evidence, and explicit lineage.

Authority decisions are published only for an explicitly closed temporal prefix. Raw invocation and revocation occurrences are retained separately from the authoritative outcome. Failed cut publication must leave no partial state. A physical-effect monitor must compare a decision with an independently trusted prefix snapshot rather than trusting a self-consistent decision object alone.

These surfaces are reference semantics. They are not a certification of a production controller.

6. Substrate interaction

Digital, analogue, classical-photonic, quantum-photonic, and quantum devices are specialized substrates. None is declared the automatic carrier of the whole C.

The substrate boundary separates observation from interpretation and decision from actuation. An effectful request is bound to:

request identity and digest
authority decision and trusted prefix
capability and version
operation/instrument profile
nonce
effect identity
receipt and audit record

For a quantum profile, the contract requires an operational measurement surface: state preparation, channel or control programme, quantum instrument or POVM, measurement outcomes, conditional update policy, noise and readout model, calibration, and uncertainty accounting.

The paper does not claim that quantum hardware is necessary for centered agency or that quantum coherence is consciousness. The first matched-control study is intended to remain classical.

7. Falsifiable matched-control programme

The relevant baseline is not a weak chatbot. It is a strong persistent-profile control with the same:

model version and sampling policy;

corpus and retrieval access;

tool set;

memory and context budgets;

perturbation stream;
 compute allocation;
 output schema and evaluator surface.

The treatment mechanisms are introduced separately before any full-bundle claim. Proposed behavioural endpoints are:

SRR - Silent Rewrite Rate
 OCA - Obligation Carryover Accuracy
 PAR - Provenance-Aware Recovery
 RCS - per-episode Reality-Correction Score
 RCG - randomized arm-level Reality-Correction Gain

These endpoints are behavioural proxies. They do not identify a hidden internal mechanism. Mechanism claims require separate conformance gates or interventions.

The programme accepts a hard null result:

If a strong matched persistent-profile control reproduces all preregistered behavioural observables and trusted conformance gates, operational equivalence must be accepted on that tested surface.

No invisible residual may be added after the fact to protect the theory.

8. What is in v0.8.1

The public release contains:

full English working paper and English normative specifications
 English public research brief
 reference Python implementations
 test and regression suites
 Windows and Linux runners
 JUnit, stdout/stderr, and typed test receipts
 concurrency checks
 16 target-specific source mutants
 six cross-analysis fixtures
 negative control against collection-only false PASS
 blind-review packet and public review forms

The reviewed execution surface reports:

v0.8 tests:	190/190
legacy v0.7 tests:	177/177
thread/process rounds:	100/20
source mutants:	16/16 killed
cross-analysis fixtures:	6/6
full-test negative control:	PASS
Windows reproduction:	PASS
sixth blind model review:	NO_CANDIDATE_BLOCKERS_FOUND

These numbers describe bounded executable evidence. They do not establish unbounded correctness, production security, empirical continuity, or subjective experience.

9. What the public is asked to test

Formal methods and distributed systems

Review temporal prefix semantics, atomic cut publication, raw-event retention, replay and fork classification, obligation lineage, liveness assumptions, and the abstraction boundary between specification and implementation.

Belief revision and decision theory

Review the distinction among revision, world update, calibration update, commitment, obligation, and policy. Challenge the SOC action universe, hard-safety rule, value-of-information policy, stopping rules, branch closure, and model-relation types.

Experimental methodology

Review construct validity, fixed opportunity sets, scorer instructions, evaluator blinding, treatment/control information parity, hierarchical analysis, equivalence margins, catastrophic-tail rules, and the separation between behavioural endpoints and mechanism evidence.

Quantum information and photonics

Review the claim boundary around coherence, no-cloning, instruments, readout, calibration, noise, postselection, classical photonics, quantum photonics, and the requirement for an end-to-end classical baseline.

Reproducibility

Run the supplied commands on a clean copy. Verify the test receipts and hashes. Report the exact platform and Python/PowerShell versions. Do not treat a changed report file as a changed frozen input.

External endpoint raters

The project seeks at least two independent raters for a pilot of SRR, OCA, PAR, and RCS. The pilot is intended to measure evaluator agreement and ambiguity, not to test the central treatment effect.

10. Claim boundary

This release does not claim:

consciousness or subjective experience
sentience or life
personhood or legal standing
AGI or entity status
empirical identity or continuity
differential witness
production security
unbounded mathematical correctness
quantum or photonic necessity

The phrase C-candidate denotes only a system that satisfies the declared architectural surface before an independent matched-control study.

11. How to contribute

Use the GitHub issue templates for formal review, conceptual review, reproducibility, or endpoint-methodology feedback. A useful report includes:

exact release and SHA-256
platform and dependency versions
file and line range
minimal scenario
observed and expected result
whether the issue is conceptual or reproduced
proposed acceptance criterion

The project welcomes negative results. A confirmed blocker creates a new version; it is not silently edited away.

12. Citation and persistent record

The version DOI reserved for this record is 10.5281/zenodo.21443105; it is registered when the Zenodo record is published. The previous DOI-bound theoretical-core release is 10.5281/zenodo.20679718.

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